

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING****1.1. Product identification**

**Product Code/Catalogue Number:** 984325  
**SDS Number:** D14765\_SDS\_Beta-Hydroxybutyrate R2 \_EN  
**Product Name** **Beta-Hydroxybutyrate R2**

**1.2. Relevant identified uses of the substance or mixture and uses advised against**

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

**1.3. Details of the supplier of the safety data sheet**

**Company** **Thermo Fisher Scientific Oy**  
Analyzers & Automation  
Clinical Diagnostics  
Ratastie 2, P.O. Box 100  
FI-01621 Vantaa, Finland  
**Telephone number** +358 10 329200  
**E-mail address** [system.support.fi@thermofisher.com](mailto:system.support.fi@thermofisher.com)

**1.4. Emergency telephone number**

CHEMTREC INTERNATIONAL +1 703-741-5970

**SECTION 2: HAZARDS IDENTIFICATION****2.1. Classification of the substance or mixture****CLP Classification - Regulation (EC) No 1272/2008**

Based on available data, the classification criteria are not met

**Classification according to EU Directives 67/548/EEC or 1999/45/EC**

Not dangerous goods.

**2.2. Label elements**

None required

**2.3. Other hazards**

No information available

**SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS***For the full text of the R-phrases and H-Statements mentioned in this Section, see Section 16.***SECTION 4: FIRST AID MEASURES****4.1. Description of first aid measures****General Advice**

If symptoms persist, call a physician.

**Inhalation**

Move to fresh air.

**Skin Contact**

Wash off with warm water and soap.

**Eye Contact**

Rinse with plenty of water.

**Ingestion**

Clean mouth with water and drink afterwards plenty of water.

**4.2. Most important symptoms and effects, both acute and delayed**

No information available.

**4.3. Indication of any immediate medical attention and special treatment needed**

Treat symptomatically.

**SECTION 5: FIREFIGHTING MEASURES****5.1. Extinguishing media****Suitable Extinguishing Media**

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

**Extinguishing media which must not be used for safety reasons**

No information available.

**5.2. Special hazards arising from the substance or mixture**

No information available.

**Hazardous Combustion Products**

None under normal use conditions.

**5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**SECTION 6: ACCIDENTAL RELEASE MEASURES****6.1. Personal precautions, protective equipment and emergency procedures**

Use personal protective equipment. Ensure adequate ventilation.

**6.2. Environmental precautions**

Prevent further leakage or spillage if safe to do so.

**6.3. Methods and material for containment and cleaning up**

Soak up with inert absorbent material.

**6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

**SECTION 7: HANDLING AND STORAGE****7.1. Precautions for safe handling**

Ensure adequate ventilation. Avoid contact with skin and eyes.

**7.2. Conditions for safe storage, including any incompatibilities**

Keep container tightly closed in a dry and well-ventilated place.

**7.3. Specific end use(s)**

Use in laboratories

**SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION****8.1. Control parameters**

Component Exposure Limits

**8.2. Exposure controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas.

**Personal protective equipment****Eye Protection**

Safety glasses with side-shields (European standard - EN 166)

**Hand Protection**

Protective gloves

| Glove material    | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|-------------------|-----------------------------------|-----------------|-------------|-----------------------|
| Disposable gloves | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Small scale/Laboratory use**

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

No information available.

**SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES****9.1. Information on basic physical and chemical properties**

|                                 |                          |                                          |
|---------------------------------|--------------------------|------------------------------------------|
| <b>Appearance</b>               | No information available |                                          |
| <b>Physical State</b>           | Liquid                   |                                          |
| <b>Odor</b>                     | No information available |                                          |
| <b>Odor Threshold</b>           | No data available        |                                          |
| <b>pH</b>                       | No data available        |                                          |
| <b>Melting Point/Range</b>      | No data available        |                                          |
| <b>Softening Point</b>          | No data available        |                                          |
| <b>Boiling Point/Range</b>      | No data available        |                                          |
| <b>Flash Point</b>              | No data available        | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>         | No data available        |                                          |
| <b>Flammability (solid,gas)</b> | No information available |                                          |
| <b>Explosion Limits</b>         | No data available        |                                          |
| <b>Vapor Pressure</b>           | No data available        |                                          |
| <b>Vapor Density</b>            | No data available        | (Air = 1.0)                              |

---

|                                                |                          |
|------------------------------------------------|--------------------------|
| <b>Specific Gravity / Density</b>              | No data available        |
| <b>Bulk Density</b>                            | No data available        |
| <b>Water Solubility</b>                        | No information available |
| <b>Solubility in other solvents</b>            | No information available |
| <b>Partition Coefficient (n-octanol/water)</b> |                          |
| <b>Autoignition Temperature</b>                | No data available        |
| <b>Decomposition Temperature</b>               | No data available        |
| <b>Viscosity</b>                               | No data available        |
| <b>Explosive Properties</b>                    | No information available |
| <b>Oxidizing Properties</b>                    | No information available |

**9.2. Other information**

No data available

**SECTION 10: STABILITY AND REACTIVITY****10.1. Reactivity**

No data available

**10.2. Chemical stability**

Stable under normal conditions

**10.3. Possibility of hazardous reactions**

No information available.

**10.4. Conditions to avoid**

None known.

**10.5. Incompatible materials**

No information available.

**10.6. Hazardous decomposition products**

None under normal use conditions.

**SECTION 11: TOXICOLOGICAL INFORMATION****11.1. Information on toxicological effects****Product Information**

No acute toxicity information is available for this product

**(a) acute toxicity;****Oral** Not Classified**Dermal** Not Classified**Inhalation** Not Classified**(b) skin corrosion/irritation;**

No data available.

**(c) serious eye damage/irritation;**

No data available.

**(d) respiratory or skin sensitization;****Respiratory**

No data available.

**Skin**

No data available.

**(e) germ cell mutagenicity;**

No data available

**(f) carcinogenicity;**

No data available

There are no known carcinogenic chemicals in this product

**(g) reproductive toxicity;**

No data available.

**(h) STOT-single exposure;**

No data available.

**(i) STOT-repeated exposure;**

No data available.

**Target Organs**

No information available.

**(j) aspiration hazard;**

No data available.

**Symptoms / effects, both acute and delayed**

No information available

**SECTION 12: ECOLOGICAL INFORMATION****12.1. Toxicity****Ecotoxicity effects**

No information available.

**12.2. Persistence and degradability**

No information available

**12.3. Bioaccumulative potential**

No information available

**12.4. Mobility in soil**

No information available

**12.5. Results of PBT and vPvB assessment**

No data available for assessment.

**12.6. Other adverse effects**

None known

**SECTION 13: DISPOSAL CONSIDERATIONS****13.1. Waste treatment methods****Waste from Residues / Unused Products**

Dispose of in accordance with local regulations.

**Contaminated Packaging**

Dispose of in accordance with local regulations.

**SECTION 14: TRANSPORT INFORMATION**

# SAFETY DATA SHEET

Beta-Hydroxybutyrate R2

Revision Date 04-Jun-2015

|                                  | IMDG/IMO      | ADR           | IATA          |
|----------------------------------|---------------|---------------|---------------|
|                                  | Not regulated | Not regulated | Not regulated |
| 14.1. UN number                  | -             | -             | -             |
| 14.2. UN proper shipping name    | -             | -             | -             |
| 14.3. Transport hazard class(es) | -             | -             | -             |
| 14.4. Packing group              | -             | -             | -             |

## 14.5. Environmental hazards

No hazards identified

## 14.6. Special precautions for user

No special precautions required

## 14.7. Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code

Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

This safety data sheet complies with the requirements of Regulation (EC) No. 1907/2006

## 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

International Inventories X = listed

### National Regulations

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has not been conducted

## SECTION 16: OTHER INFORMATION

### Legend

CAS - Chemical Abstracts Service

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

WEL - Workplace Exposure Limit

ACGIH - American Conference of Governmental Industrial Hygienists

DNEL - Derived No Effect Level

RPE - Respiratory Protective Equipment

LC50 - Lethal Concentration 50%

NOEC - No Observed Effect Concentration

PBT - Persistent, Bioaccumulative, Toxic

ADR - European Agreement Concerning the International Carriage of Dangerous Goods by Road

IMO/IMDG - International Maritime Organization/International Maritime Dangerous Goods Code

OECD - Organisation for Economic Co-operation and Development

BCF - Bioconcentration factor

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

NZIoC - New Zealand Inventory of Chemicals

TWA - Time Weighted Average

IARC - International Agency for Research on Cancer

PNEC - Predicted No Effect Concentration

LD50 - Lethal Dose 50%

EC50 - Effective Concentration 50%

POW - Partition coefficient Octanol:Water

vPvB - very Persistent, very Bioaccumulative

ICAO/IATA - International Civil Aviation Organization/International Air Transport Association

MARPOL - International Convention for the Prevention of Pollution from Ships

ATE - Acute Toxicity Estimate

VOC - Volatile Organic Compounds

### Key literature references and sources for data

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

# SAFETY DATA SHEET

Beta-Hydroxybutyrate R2

Revision Date 04-Jun-2015

---

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

|                     |                       |
|---------------------|-----------------------|
| Version             | 1                     |
| Revision Date       | 04-Jun-2015           |
| Reason for revision | Update to CLP Format. |

## Disclaimer

The information provided on this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guide for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered as a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other material or in any process, unless specified in the text.